

ELEC4631 Solutions Tutorial 10

1. $R(s) = s^2 + 6s + 10$. Using Ackermann's Formula gives

$$K = [\ 0 \quad 1 \] \mathcal{C}^{-1} R(A) = [\ 0.4545 \quad -1.5455 \].$$

Let $r(t) = c1(t)$, where $1(t)$ is the unit step function. Then when $t \geq 0$, we have that $r(t) = c$, so the control law is $u(t) = -Kx(t) + Nc$ when $t \geq 0$. Following Lecture 13, $N = u_{ss} + Kx_{ss}$ where x_{ss} and u_{ss} solve the equation:

$$\begin{bmatrix} -5 & 3 & 1 \\ -2 & 1 & -1 \\ 4 & 3 & 2 \end{bmatrix} \begin{bmatrix} x_{ss} \\ u_{ss} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Solving gives

$$x_{ss} = \begin{bmatrix} 0.1143 \\ 0.2 \end{bmatrix}, \quad u_{ss} = -0.0286.$$

So, $N = u_{ss} + Kx_{ss} = -0.2857$. Therefore, the control law is $u(t) = -[\ 0.4545 \quad -1.5455 \]x(t) - 0.2857r(t)$.

2. (a) $\gamma(s) = \det(sI - A + BK + LC - \beta K)$. Let $X = B - \beta$ then $\gamma(s) = \det(sI - (A - LC) + XK)$. If $(K, A - LC)$ is an observable pair then using Ackermann's Formula for an observer

$$X = \gamma(A - LC) \begin{bmatrix} K \\ K(A - LC) \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix},$$

with $\beta = B - X$.

(b) $A - LC = \begin{bmatrix} -5 & 3 \\ -6 & -2 \end{bmatrix}$, $\gamma(A - LC) = (A - LC)^2 + 6(A - LC) + 25I_2 = \begin{bmatrix} 2 & -3 \\ 6 & -1 \end{bmatrix}$,
 $X = \begin{bmatrix} 0.0476 \\ 0.5238 \end{bmatrix}$ and $\beta = B - X = \begin{bmatrix} 0.9524 \\ -1.5238 \end{bmatrix}$.

3. (a) The input is $r(t) = c1(t)$, the Laplace transform is $R(s) = 1/s$. Let the output be $y(t)$. Then the Laplace transform of the output is $Y(s) = H(s)R(s) = \frac{cH(s)}{s}$. We want $\lim_{t \rightarrow \infty} y(t) = c$. But we also have $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} sH(s)R(s) = \lim_{s \rightarrow 0} cH(s) = cH(0)$. Therefore, we must have $H(0) = 1$ (the dc gain of the transfer function is 1).
- (b) $H(0) = \frac{3 \times 7 \times 2}{1 \times 3 \times 14} = 1$.
- (c) $e_{ss} = \frac{1}{K^{\text{Truxal}}} = \sum \frac{1}{z_k} - \sum \frac{1}{p_k} = 16/21$.
- (d)

$$G(s) = \frac{H(s)}{1 - H(s)} = \frac{3(s+7)(s+2)}{s(s^2 + 15s + 32)},$$

so $G(s)$ is Type 1, as it should be.

- (e) $K_v = \lim_{s \rightarrow 0} sG(s) = \frac{21}{16}$. $K^{\text{Truxal}} = 1/e_{ss} = 21/16$. Therefore, $K_v = K_v^{\text{Truxal}}$, as expected.
4. (a) The controllability matrix is

$$\mathcal{C} = \begin{bmatrix} -1 & -2 \\ 1 & 3 \end{bmatrix},$$

which is invertible, so the system is controllable. We check the following matrix,

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ -2 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

which is clearly not invertible since the first and third columns are linearly dependent. For this problem, one cannot solve for the equation

$$\begin{bmatrix} 2 & 0 & -1 \\ -2 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_{ss} \\ u_{ss} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

to get x_{ss} and u_{ss} . This is true in general for the scenario discussed in slide #13 of Lecture 13. (**Note:** the uninvertibility of a matrix on the left side of a linear equation does not necessarily imply that the equation does not have a solution, the other possibility is that there is more than one solution. It depends on what is on the right hand side of the equation. In this case, the right hand side is such that there is indeed no solution to this equation. The reason is that the second row on the left is a certain linear combination of the first and third rows, but the second element on the right hand side is not the same linear combination of the first and third elements. Hence, the equation is not consistent and has no solution. Think about it, this is again just basic linear algebraic facts).

(b) The transfer function is

$$\begin{aligned}
H(s) &= C(sI - A)^{-1}B + D, \\
&= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} s-2 & 0 \\ 2 & s-1 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \\
&= \frac{s}{(s-1)(s-2)}.
\end{aligned}$$

What is distinctive about this transfer function is that it has a zero at $s = 0$. It is this zero at $s = 0$ that is causing the problem. Now, suppose that K is any stabilizing feedback gain (when the system is controllable such a stabilising feedback gain always exists. If the system is not controllable a stabilising gain may or may not exist, depending on the system). Consider a control law of the form $u = -Kx + Nr$, where r is any step function of any amplitude and $N \neq 0$. The system with this control signal is then

$$\dot{x} = (A - BK)x + BNr, \quad (1)$$

$$y = Cx. \quad (2)$$

The polynomial for the zeros of the transfer function from r to y under this control law is:

$$z_{cl}(s) = \det \begin{bmatrix} sI - A + BK & -BN \\ C & 0 \end{bmatrix}. \quad (3)$$

Now, do the following column operations: multiply the last column of $z_{cl}(s)$, $\begin{bmatrix} -BN \\ 0 \end{bmatrix}$, by K/N and add this to the first two columns $\begin{bmatrix} sI - A + BK \\ C \end{bmatrix}$. These column operations do not change the determinant, so one ends up with the following:

$$z_{cl}(s) = \det \begin{bmatrix} sI - A + BK & -BN \\ C & 0 \end{bmatrix}, \quad (4)$$

$$= \det \begin{bmatrix} sI - A & -BN \\ C & 0 \end{bmatrix}, \quad (5)$$

$$= N \det \begin{bmatrix} sI - A & -B \\ C & 0 \end{bmatrix}, \quad (6)$$

But

$$z_{ol}(s) = \begin{bmatrix} sI - A & -B \\ C & 0 \end{bmatrix} = s,$$

since this is the polynomial for the zeros of the original plant under no control. So, we have that the closed-loop zero polynomial is

$$z_{cl}(s) = Ns.$$

That is, the closed-loop transfer function from r to y still has a zero at $s = 0$! It is not possible at all to get rid of this under the control law $u = -Kx + Nr$ and this is the reason why this system cannot track a step input under any control law of this kind.

(a) The transfer function is:

$$\begin{aligned} G(s) &= C(sI - A)^{-1}B + D, \\ &= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} s & 0 \\ -1 & s+1 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 4 \end{bmatrix}, \\ &= \begin{bmatrix} 0 & 1 \end{bmatrix} \frac{\begin{bmatrix} s+1 & 0 \\ 1 & s \end{bmatrix}}{s(s+1)} \begin{bmatrix} 6 \\ 4 \end{bmatrix}, \\ &= \frac{4s+6}{s(s+1)}, \end{aligned}$$

(b) The system has one integrator, so it is Type 1.

(c) We want two dominant closed-loop poles to be at $p_1 = -4 + 3j$ and $p_2 = -4 - 3j$. To do this we can set the controller gain K by Ackermann's formula and get that $K = \begin{bmatrix} -4.8333 & 9 \end{bmatrix}$. Let the observer poles be p_3, p_4 . Since $G(s)$ has a zeros at $-3/2$, the closed-loop zeros will be the zeros of $\gamma(s)$ together with $-3/2$. That is, if z_1 and z_2 that are roots of $\gamma(s)$ then the closed-loop zeros will be $z_1, z_2, -3/2$. To achieve the desired K_v , we need to set z_1, z_2, p_3 , and p_4 such that the Truxal's formula is satisfied:

$$\frac{1}{K_v} = \frac{1}{z_1} + \frac{1}{z_2} - \frac{2}{3} - \frac{1}{p_1} - \frac{1}{p_2} - \frac{1}{p_3} - \frac{1}{p_4}.$$

Since p_1 and p_2 have been chosen and $K_v = 20$, this is one equation in four variables z_1, z_2, p_3, p_4 . Hence it can have more than one solution. As discussed in the lecture notes, to minimize their effect on the closed-loop dynamics we choose z_1 and p_3 in the LHP such that they are close to each other, and also z_2 and p_4 in the LHP to be close to each other. Apart from that there are no other constraints, so to simplify the task we can set $z_1 = z_2 = z$ and $p_3 = p_4 = p$, giving the equation:

$$\frac{1}{K_v} + \frac{1}{p_1} + \frac{1}{p_2} + \frac{2}{3} = \frac{2}{z} - \frac{2}{p}.$$

This is still one equation in two variables z and p . We can set z , and then this will give us p . Try $z = -0.4$, solving the equation above gives $p = -0.3706$. So, we get $z_1 = z_2 = -0.4$ and $p_3 = p_4 = -0.3706$. Note that z_1 is close to p_3 and z_2 close to p_4 as desired, so this solution is acceptable. From p_3 and p_4 we can compute the observer gain as $L = \begin{bmatrix} 0.1373 \\ -0.2588 \end{bmatrix}$.

So, we get that $\gamma(s) = (s-z_1)(s-z_2) = s^2 + 0.8s + 0.16$. From the formula obtained for Problem 2 above we can compute $X = \begin{bmatrix} -0.0032 \\ -0.0082 \end{bmatrix}$ and $\beta = \begin{bmatrix} 5.9968 \\ 3.9918 \end{bmatrix}$.

Finally from the formula in lectures, to get a closed-loop DC gain of 1, $N = 3.5767$ and $M = \beta N = \begin{bmatrix} 21.4487 \\ 14.2772 \end{bmatrix}$.

- (d) As mentioned earlier, the closed-loop system zeros will be the roots of $\gamma(s)$ together with the zeros of $H(s)$. Since $H(s)$ has $-3/2$ as a zero, the closed-loop zeros are therefore $z_1 = -0.4$, $z_2 = -0.4$, $-3/2$. The closed-loop poles are p_1 and p_2 (set by the gain K) and the poles p_3 and p_4 (set by the observer). Thus the closed-loop transfer function is of the form:

$$H_{cl}(s) = K_{cl} \frac{(s + 3/2)\gamma(s)}{(s - p_1)(s - p_2)(s - p_3)(s - p_4)},$$

for some real constant K_{cl} . The value of K_{cl} can be determined from the fact that $H_{cl}(0) = 1$ (the closed-loop DC-gain is 1). This gives

$$K_{cl} = \frac{(-p_1)(-p_2)(-p_3)(-p_4)}{\gamma(0)(3/2)} = 14.3067.$$

- (e) Using $H_{cl}(s)$, one can plot the step and ramp response on Matlab using the `lsim` function.